

Predicting Academic Performance using Gaming Habits and AI Usage

This project is basically about trying to predict how students perform in school, like their grades or GPA changes, using different types of data, so for one dataset I used gaming habits, like how much someone plays, sleeps, studies, stuff like that, and for the other dataset I used AI usage, like how often students use tools like ChatGPT and how that might affect their GPA, so overall the goal was to see if machine learning models can actually learn patterns from this kind of data and make good predictions.

The first dataset I used was the Gaming Academic Performance dataset, and this one has things like gaming hours, study hours, sleep, attendance, and other lifestyle stuff, and the target for this dataset was the student's grades, which made sense because all the features are kind of related to academic performance.

The second dataset was AI Student Life, and this one includes things like what AI tools students use, how often they use them, and their GPA before and after using AI, so for this one I created a new target called GPA change, which is just GPA after minus GPA before, so I could try to predict how AI usage impacts performance.

For both datasets I used a few different machine learning models, including Linear Regression, Random Forest, and KNN (and Decision Tree earlier), just to compare how different types of models perform, and I used pipelines so that all the preprocessing like scaling and encoding happens automatically, which makes things cleaner and easier to manage.

I also split the data into training and testing sets, so I could evaluate how well the models actually generalize, and then later I used GridSearchCV to tune the Random Forest model, by testing different values like number of trees, max depth, and split size, to try and improve performance.

For Dataset 1 (Gaming Academic Performance), the results were actually really strong, the tuned Random Forest model had an R^2 of about 0.92, which means it explained most of the variation in grades, and the error values (MAE and RMSE) were also pretty reasonable, so overall this dataset worked really well and the model was able to learn clear patterns.

For Dataset 2 (AI Student Life), the results were not good, the R^2 was negative (around -0.02), which basically means the model did worse than just guessing the

average, and even after tuning the model didn't improve much, so this dataset was much harder to predict from.

For Dataset 1, the results match pretty well with the study by Mahmud et al. (2023), which found that students who spend more time gaming tend to have lower academic performance, less study time, and worse habits overall, so this supports the idea that the features in the dataset are actually useful for predicting grades.

For Dataset 2, the results are kind of different from the study by Wecks et al. (2024), which found that students using AI tools scored lower on exams on average, but in my case the model couldn't really find strong patterns, and I think that's probably because the dataset is smaller and GPA change is harder to predict, since it depends on a lot of other factors that aren't included.

Overall, this project showed that machine learning can work really well when the dataset has strong relationships between features and the target, like in the gaming dataset, but when the data is more complex or missing important factors, like the AI dataset, the models struggle a lot, even after tuning, so the main takeaway is that the quality and structure of the data matters just as much as the model itself, maybe even more, and that not every dataset will give good predictions even if you use good algorithms.